



Sobha Realty Delayed Transaction & Revenue Exposure Analysis

Milestone 2

Group Number: 08

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1. Problem Statement

Sobha Realty has a large portfolio of property transactions across multiple projects and property types. A significant number of transactions in the available 2018-2023 dataset are classified as delayed. This project analyzes where these delayed transactions are concentrated across projects, property types, and years, quantifies the revenue associated with them, and compares delay rates across projects to identify areas that may warrant greater management attention. The analysis is limited to patterns observable in the transaction data and does not attempt to determine the underlying causes of delays.

A transaction is treated as delayed when its ProcedureNameEn field carries one of the following status labels: Complete Delayed Sell, Delayed Sell, Delayed Mortgage, Grant on Delayed Sell, Delayed Sell Lease to Own Registration.

Milestone 1 answered 7 foundational questions about overall delay volume, yearly trend, top delayed projects, property types, and revenue exposure. Milestone 2 builds on this by combining dimensions together, ranking projects on more than one measure, and looking at additional factors such as property size and buyer nationality.

2. Database

The dataset was obtained from an actual client business (identity kept confidential for privacy). It originally had 105,283 rows covering property transactions from 2018 to 2023 across 116 projects.

Before analysis, the data was cleaned in Excel:

- Duplicate rows were removed
- Rows with missing or invalid project names were removed
- Rows with a price of 0 or blank were removed
- Columns with personal buyer/seller information (name, mobile number, ID numbers) were deleted for privacy reasons

After cleaning, the analysis was run only on rows where the party type is 'Buyer', to avoid counting the same transaction twice (each transaction otherwise appears once for the buyer and once for the seller).

2. Data Validation

Before running the Milestone 2 queries, the same cleaned dataset from Milestone 1 (sobha_cleaned.csv) was reused so no additional cleaning was required. For queries that group by Project or by combinations such as Project + Property Type, a minimum transaction count filter (HAVING COUNT(*) >= 30, or >= 100 for the best-performer comparison) was applied. This was done to avoid a project or combination with only 1-2 transactions appearing at the top of a ranking purely by chance.

3. Tools Used

- SQL — DataLab (DataCamp) using DuckDB
- Microsoft Excel — data cleaning and preparation

4. Questions Answered

1. Which project + property type combination has the highest number of delayed transactions?
2. Which project has both high sales volume and a low delay rate (the best-performing project)?
3. Which projects have delays in every single year, not just once?
4. Do smaller properties get delayed more than bigger ones?
5. Which nationality's buyers are most affected by delayed transactions?
6. How has the delay percentage changed year by year?
7. Which property type has grown the fastest in transaction volume over the years?
8. What are the 10 most expensive delayed transactions?

5. SQL Techniques Used

- GROUP BY with two columns together (Project + Property Type)
- Multi-column ORDER BY (sorting by transaction volume, then by delay rate)
- CASE WHEN used to create new business categories (Small/Medium/Large property size groups) - this is different from Milestone 1, where CASE WHEN was only used to flag delayed vs. not delayed.
- A CTE (WITH clause) to calculate delays per project per year, then check how many distinct years each project appears in
- HAVING clauses to filter out low-volume groups before ranking

Note: on JOINS: the dataset consists of a single table, so there was no second table to JOIN against. A CTE was used in Question 3 instead, to demonstrate a similar level of multi-step SQL logic.

6. Analysis & Findings

Q1. Which Project +Property type combination has the highest number of delayed transactions?

Why this query: goes one level deeper than Milestone 1 - instead of looking at project or property type separately, it looks at specific combinations, which is more actionable for management.

```
SELECT
    Project,
    PropertyTypeEn,
    COUNT(*) AS total_transactions,
    SUM(CASE WHEN ProcedureNameEn IN
        ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
         'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
        THEN 1 ELSE 0 END) AS delayed_transactions
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer'
GROUP BY Project, PropertyTypeEn
HAVING COUNT(*) >= 30
ORDER BY delayed_transactions DESC
LIMIT 10;
```

Output:

What we found : Sobha Creek vistas “flat” units have the highest number of delayed transactions .

Number \ Evidence : out of 2135 total transactions , 1067 were delayed [about 50 %]

What It Means : This shows that this specific project property combination is consistently facing late deliveries . it's a red flag of buyers considering this project and it raises questions about the developers construction \ delivery capacity for this particular unit type.

Q2. Which Project has both high sales volume and low delay rate [the best performing project]?

Why this query: run to independently confirm the total transaction count used as the denominator for the delay percentage above.

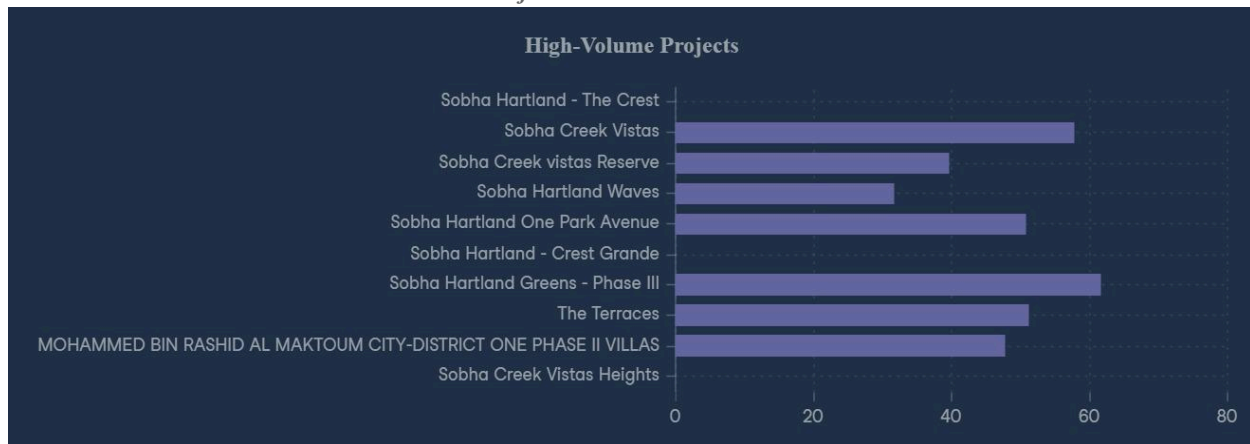
```
SELECT
    Project,
    COUNT(*) AS total_transactions,
    ROUND(
        SUM(CASE WHEN ProcedureNameEn IN
            ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
```

```

        'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
    THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2
) AS delay_percentage
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer'
GROUP BY Project
HAVING COUNT(*) >= 100
ORDER BY total_transactions DESC, delay_percentage ASC
LIMIT 10;

```

Project-Wise Sales Volume



Output:

What we found : Sobha Hartland The Crest is the best performing project it has high sales volume and zero delays .

Number \ Evidence : 3472 total transactions with 0 % delay rate .

What it Means : This proves that on time delivery is possible even with high demand . It also shows that delays are not just caused by market pressure or sales volume , they depend on how well a project is managed . Compare this to Sobha Creek Vistas [similar volume 3127 but 57.85 % delayed] same developer different outcome which highlights the difference in execution between projects .

Q3. Which Project has delays in every single year not just once ?

Why this query: groups transactions by year to see whether delays are getting better or worse over time, not just what the current overall rate is.

```

WITH yearly_delays AS (
SELECT
    Project,
    YEAR(
        COALESCE(
            TRY_STRPTIME(Regis, '%d/%m/%Y'),
            TRY_STRPTIME(Regis, '%d-%b-%y')
        )
    ) AS transaction_year,
    SUM(CASE WHEN ProcedureNameEn IN
        ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
        'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
        THEN 1 ELSE 0 END) AS delayed_count
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer'

```

```

GROUP BY Project, transaction_year
)
SELECT
    Project,
    COUNT(DISTINCT transaction_year) AS years_with_delays
FROM yearly_delays
WHERE delayed_count > 0
GROUP BY Project
HAVING COUNT(DISTINCT transaction_year) >= 5
ORDER BY years_with_delays DESC
LIMIT 10;

```

Outcome :

What we found : some projects have had delays every single year on record , not just once this is a chronic problem , not a one off issue .

Number\ Evidence : projects like Muhammad Bin Rashid al Makhtoum City, District One Phase 11 Villas , Sobha Hartland Villas phase 1 and district 1 phase recorded delays in 6 out of 6 years.

What it Means : A delay in one year could be a coincidence but a delay every single year points to a systematic issue such as poor planning , funding problems, or ongoing management issues . These projects falls into a high risk category for future buyers .

Q4. Do smaller properties get delayed more than bigger ones ?

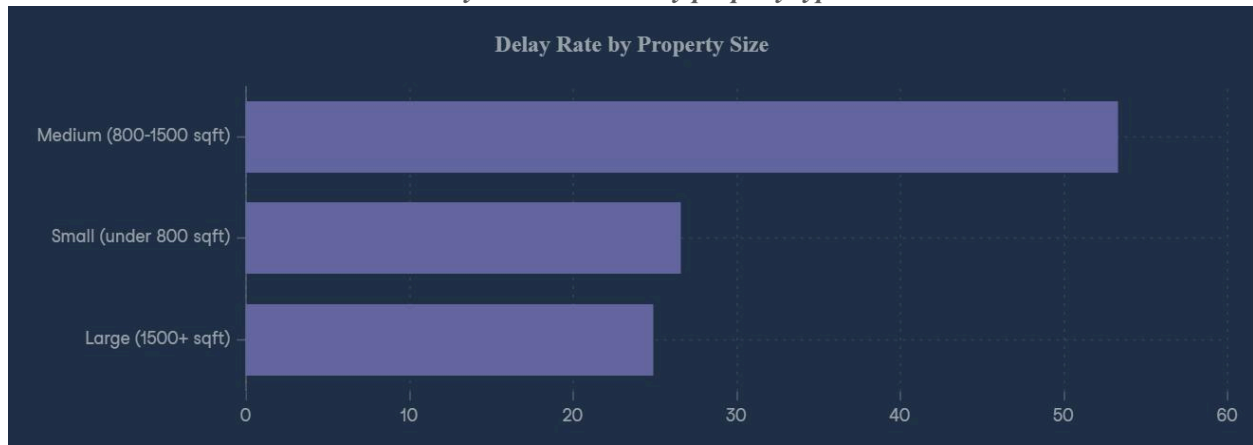
Why this query: ranks projects by raw delayed-transaction count to identify where management attention is most urgently needed in absolute terms.

```

SELECT
    CASE
        WHEN Size < 800 THEN 'Small (under 800 sqft)'
        WHEN Size >= 800 AND Size < 1500 THEN 'Medium (800-1500 sqft)'
        WHEN Size >= 1500 THEN 'Large (1500+ sqft)'
        ELSE 'Unknown'
    END AS size_category,
    COUNT(*) AS total_transactions,
    SUM(CASE WHEN ProcedureNameEn IN
        ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
        'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
        THEN 1 ELSE 0 END) AS delayed_transactions,
    ROUND(
        SUM(CASE WHEN ProcedureNameEn IN
            ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
            'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
            THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2
        ) AS delay_percentage
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer' AND Size IS NOT NULL
GROUP BY size_category
ORDER BY delay_percentage DESC;

```

Delayed transactions by property type



Outcome :

What we found : There is not a simple 'smaller property= more delays' pattern. Infact medium sized properties have the highest delay rates not the small ones.

Number \ Evidence: Medium [800-1500 sqft]:53.34% delay rate \small[under 800 sqft] ;26.61 % \Large [1500+sqft]: 24.91%

What it Means : Our initial Assumptions was wrong size and delay don't have a simple straight line relationship . Medium sized properties are delayed more , possibly because these mid range projects face more complex construction challenges or a bigger demand supply mismatch . Large properties tend to be better managed likely because they are backed by better funded developers.

Q5: Which Nationality buyers are most affected by delayed transactions ?

Why this query: breaks delays down by property type, and calculates delay_percentage (not just raw count) so that small but high-risk categories are not hidden behind larger ones.

```
SELECT
    CountryNameEn,
    COUNT(*) AS total_transactions,
    SUM(CASE WHEN ProcedureNameEn IN
        ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
        'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
        THEN 1 ELSE 0 END) AS delayed_transactions,
    ROUND(
        SUM(CASE WHEN ProcedureNameEn IN
            ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
            'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
            THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2
    ) AS delay_percentage
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer' AND CountryNameEn IS NOT NULL
GROUP BY CountryNameEn
HAVING COUNT(*) >= 50
ORDER BY delay_percentage DESC
LIMIT 10;
```

Top 10 Countries Buyer Affected Of Delayed Transactions



Outcome:

What we found : Buyers from Gabon are the most affected by delayed transactions in percentage terms.

Numbers/ Evidence : Gabon 64 out of 99 transactions delayed= 64.65%delay rate the highest of any nationality . Followed by Tanzania which is (62.5%), China (54.35%)and Lebanon (45.32%)

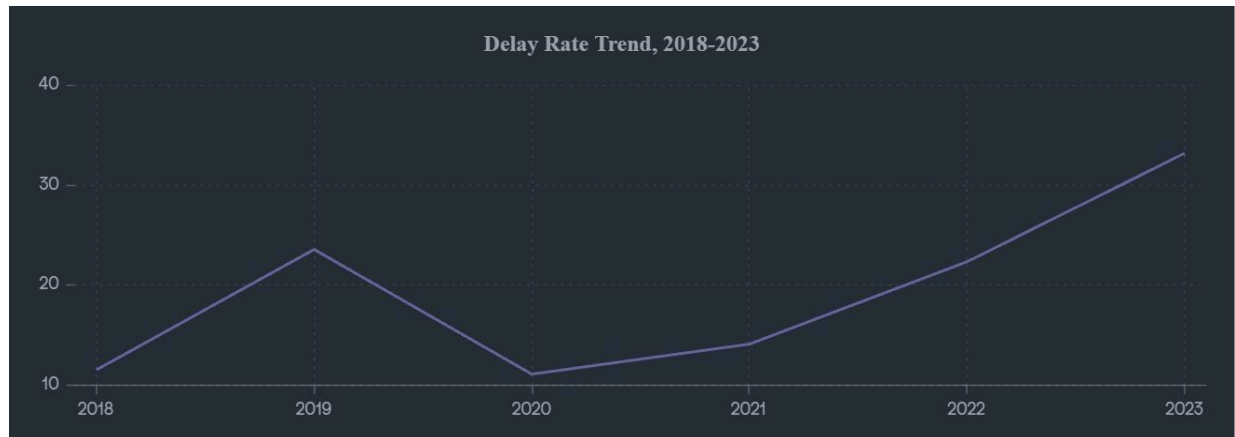
What It Means : The pattern suggests that certain nationality groups may be choosing riskier projects /developers or may have less access to information or due diligence resources before investing . This is an important insight for designing buyer protection policies targeted awareness or regulation for specific buyer segments may be needed.

Q6. How has the delay percentage changed year by year ?

Why this query: compares average transaction value for delayed vs. normal transactions to see whether higher-value deals are more or less likely to be delayed.

```
SELECT
    YEAR(
        COALESCE (
            TRY_STRPTIME (Regis, '%d/%m/%Y'),
            TRY_STRPTIME (Regis, '%d-%b-%y')
        )
    ) AS transaction_year,
    COUNT(*) AS total_transactions,
    SUM(CASE WHEN ProcedureNameEn IN
        ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
        'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
        THEN 1 ELSE 0 END) AS delayed_transactions,
    ROUND(
        SUM(CASE WHEN ProcedureNameEn IN
            ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
            'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
            THEN 1 ELSE 0 END) * 100.0 / COUNT(*), 2
    ) AS delay_percentage
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer'
GROUP BY 1
ORDER BY 1;
```

Delay percentage change trend by year



Output:

What We Found ; The delay percentage has not steadily increased or decreased over the years , it has fluctuated up and down .

Numbers / Evidence : 2018 :11.45%----2019 : 23.51 %----2020 ; 11.02%(drop) ----2021 ; 14.03% (slight increase).

What It Means: There is no steady improving or worsening trend over time . Market conditions regulations or external events are effecting each year differently . It would be misleading to say things are simply getting better or worse overall . We need to look at year specific factors(what caused the spike 2019) .

Q7. Which property type has grown the fastest in transactions volume over the years ?

Why this query: looks at revenue exposure rather than transaction count, and filters out very low-volume projects (fewer than 30 transactions) so a single delayed sale in a tiny project doesn't distort the ranking.

```
SELECT
    PropertyTypeEn,
    YEAR (
        COALESCE (
            TRY_STRPTIME (Regis, '%d/%m/%Y'),
            TRY_STRPTIME (Regis, '%d-%b-%Y')
        )
    ) AS transaction_year,
    COUNT(*) AS total_transactions
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer' AND PropertyTypeEn IN ('Flat', 'Unit',
'Villa')
GROUP BY PropertyTypeEn, transaction_year
ORDER BY PropertyTypeEn, transaction_year;
```

Output:

What We Found : Flats have grown the fastest in transaction volume over the years .

Numbers/ Evidence: Flats:440 (2018)----1613(2020)----3044(2021)---roughly 7x growth in 3 years.

What It Means : This shows a strong market shift towards apartments /flats likely driven by affordability , urbanization, or investor preference . For developers and policy makers this is clear signals that future supply planning should prioritize flats .

Q8. What are the 10 most expensive delayed transactions ?

Why this query: This query identifies the 10 most expensive delayed transactions by filtering delayed records, sorting them by transaction value in descending order, and limiting the results to the top 10. It helps us understand which individual delayed transactions have the highest financial value and where the greatest revenue exposure is concentrated.

```
SELECT
    Project,
```



```

PropertyTypeEn,
TRY_CAST(ProcedureValue AS DOUBLE) AS price,
ProcedureNameEn
FROM 'sobha_cleaned.csv'
WHERE ProcedurePartyTypeNameEn = 'Buyer'
AND ProcedureNameEn IN
    ('Complete Delayed Sell', 'Delayed Sell', 'Delayed Mortgage',
    'Grant on Delayed Sell', 'Delayed Sell Lease to Own Registration')
ORDER BY price DESC
LIMIT 10;

```

Output:

What We Found : The Most expensive delayed transactions are dominated by a Single project cluster , Mohammed Bin Rashid Al Makhtoum City District one Phase 2 villas .

Numbers/ Evidence: Top entries include AED 95613985 (villas), AED 91952320 (villas) , and AED 90000000(Both land and villa) all from this same project .

What It Means : This shows that delays are not limited to budget /affordable housing even high value luxury properties face delays and the financial risk is huge since transaction amounts runs into the tens of millions. This is notable because its generally assumed that premium projects from big developers are more reliable this findings challenge that assumptions .

8. Limitations

This analysis is based only on the transaction status field in the dataset, which labels certain transactions as delayed. The dataset does not include an expected or actual handover date, so this project cannot say why a transaction was delayed or confirm that it reflects a late property handover to the buyer. The findings only describe patterns in the recorded transaction data.

The study has some limitations that should be considered when interpreting the results. First, the analysis was based on the available transaction data, which may not include every factor that can contribute to delays. Some important factors, such as changes in buyer decisions, internal management issues, unexpected legal problems, or external market conditions, may not have been fully recorded in the dataset.

Second, the accuracy of the findings depends on the quality and completeness of the available data. Missing values, inconsistent records, or incorrect dates may affect the calculated delay periods and other statistical results.

Another limitation is that the study mainly focuses on identifying patterns and factors associated with transaction delays. Therefore, the results may show relationships between variables but may not always prove that one factor directly causes a delay.

Finally, the findings may be specific to the projects and time period included in the study. Therefore, they may not be directly applicable to every real-estate project or market without further analysis.

9. What We Learned

This project helped us understand how to move from a large, messy dataset to a clear business question. Through this analysis, we learned that real-estate transaction delays can vary across different projects, property types, and years, rather than following one common pattern. We also learned the importance of maintaining accurate and organized data. Transaction dates, project information, payment details, documentation, and completion status can be important for understanding transaction performance and identifying where further investigation may be needed. The study helped us understand how SQL can be used to identify trends and patterns that may not be obvious from individual transactions. By comparing delay rates across different groups, we could see how transaction performance varied and identify areas that may require greater management attention.

Most importantly, we learned that data-driven decision-making can help real-estate organizations monitor projects more effectively, improve operational efficiency, and make better-informed decisions. This study gave us practical experience in using real-world data to investigate a business problem, develop meaningful conclusions, and use SQL to answer business questions step by step. We also learned that raw counts alone can be misleading. Calculating percentages and revenue exposure gave us a clearer picture of where the potential risk was concentrated in the available data.